

Artifact 26 — Harness Audit Grader

Execution · Stage D · Pre-ECC-Build Specification Audit

DOCUMENT DETAILS

Feature	Home-Care-AI Scheduling Platform — Pre-ECC-Build Specification Audit	Audit Date	2026-03-28
Auditor	Harness Audit Grader (pm-execution skill)	Framework	Australian Privacy Act 1988 (APP) — HIPAA architecture as design floor
PRD	Artifact 21 (v1.0)	Logic Spec	Artifact 23 (v1.0)
User Stories	Artifact 24 (v1.0)	Synth Data	Artifact 25 (v1.0) — clients.json, incidents.json, expected_audit.sql

Schema Naming: Implementation class is `APPAuditLogEntry`. `HIPAAAuditLogEntry` is the field-structure design floor only — no production code may use the HIPAA name. See CLAUDE.md Article IX Naming Convention.

Table of Contents

Table of Contents 2

Overall Verdict 4

1. Security Debt Findings 5

 SD-02 — Authentication & Authorization (Deferred by Design) 5

 SD-03 — Prompt Injection (CRIT-02 Satisfied) 5

 SD-04 — RAG / Vector Store Isolation 6

 SD-05 — Secrets & Credentials 6

2. Requirement Drift Findings 7

3. Low-Severity Findings 9

4. Passed Checks 10

5. Agent Fix Queue 12

 FIX-001 [CRITICAL] SD-01B-001 12

 FIX-002 [HIGH] RD-02-001 12

 FIX-003 [HIGH] RD-04-001 + RD-06-001 13

 FIX-004 [HIGH] CP-01-001 13

6. Regression Test Recommendations 15

7. Pre-Sprint Gate Status 16

8. Spec Coverage Map 17

Overall Verdict

AUDIT VERDICT

CONDITIONAL PASS

								PASS	16 / 24 total checks
CRITICAL	1	HIGH	3	MEDIUM	2	LOW	2		

Merge gate: SD-01B — BLOCKED until CRITICAL finding resolved. ECC build MAY commence for the 6 implemented scenarios. ECC build MUST NOT ship to production until:

1. Security Debt Findings

ID	Check	Severity	File Path	Line(s)	Finding	Fix Instruction
SD-01B-001	Audit Log Completeness — Event Coverage	CRITICAL	Execution/synthetic-data/incidents.json	all	5 canonical events from the Artifact 23 §8 31-event registry and 2 logic execution paths have no synthetic test scenario. ECC has no ground truth to validate against for: COORDINATOR_OVERRIDE (Gate 7b), E3_GATE_BLOCKED, DUPLICATE_VACANCY_SUPPRESSED (Gate 0), CARER_NOTIFICATION_FAILED, CLIENT_NOTIFICATION_UNAVAILABLE, P-8=true FAMILIARITY_BINAR Y_FLOOR path, EC-03 partial SPP path (0 < P-10 < 0.80).	See FIX-001 — add SYN-06, SYN-07, SYN-08 scenarios with corresponding APPAuditLogEntry ground truth rows to expected_audit.sql.
SD-01B-002	Unit Test Fixture Files	MEDIUM	tests/	—	No tests/ directory exists in the repository. Artifact 23 §11 Integration Test Requirements reference unit test fixtures for hash-chain validation, Gate 4 filter, and familiarity score computation. Without these, the SD-01B merge gate query in expected_audit.sql is the only automated guard — and it only runs at integration level.	Create tests/unit/ with fixtures for: (a) familiarity_score formula (visit_count 0–6), (b) Gate 4 P-3=KNOWN_CARERS_ONLY filter, (c) hash-chain entry_hash == SHA-256(all fields). See TEST-001 through TEST-003.
CP-01-001	SMS Copy Approval Gate (B1-B)	HIGH	config/sms_templates.py	45–46	APPROVED_BY = None and APPROVED_DATE = None. The harness-audit-grader CP-01 check asserts both are non-null before production deploy. assert_copy_approved() correctly blocks in production mode and logs WARNING in H OMECARE_AI_TEST_MODE=true — the gate mechanism is implemented correctly but the approval itself has not been obtained.	Process action: Privacy Counsel reviews ACT_C_01, ACT_C_02_VARIANT_A/B/C, ACT_P_01_VARIANT_A/B, ACT_F_01_VARIANT_A/B. On approval, set APPROVED_BY = "Privacy Counsel — [Full Name]" and APPROVED_DATE = "YYYY-MM-DD". Also set COORDINATOR_PROXY_REVIEW_BY = "Angela CC-001" on coordinator clarity review. This is a process action — not an ECC code fix.

SD-02

Authentication & Authorization (Deferred by Design)

No authentication, RBAC, or JWT specification is present in Artifact 21 PRD or Artifact 23 Logic Spec. This is not filed as a HIGH finding because v1 is explicitly scoped as a coordinator-desktop tool with no public API endpoints and no multi-tenant access. However, the spec does not document this deferral explicitly. **LOW finding filed below (IS-AUTH-001).**

SD-03

Prompt Injection (CRIT-02 Satisfied)

config/sms_templates.py enforces CRIT-02 at the structural level: template.replace("{variable}", sanitized_value) sequential substitution only. No format() calls with unvalidated input, no f-strings with user data, no LLM in the SMS rendering path. get_template()

raises ValueError on any None template key, preventing accidental rendering of the unreachable ACT_P_01_VARIANT_C branch. **PASS.**

SD-04

RAG / Vector Store Isolation

Not applicable in v1. No RAG component, no vector store, no embeddings pipeline. Scheduling logic is rule-based (Artifact 23 §2). **PASS.**

SD-05

Secrets & Credentials

No hardcoded API keys, connection strings, or credentials found in any file. APPROVED_BY = None in config/sms_templates.py is an intentional approval placeholder, not a credential. AWS credentials, SMS gateway keys, and AlayaCare API tokens are correctly documented as environment variable references in Artifact 23 §9 Infrastructure. HOMECARE_AI_TEST_MODE and HOMECARE_AI_AX02_STATUS are environment variable gates — correct pattern. **PASS.** Minor gap filed as LOW (see AX-02-001 below).

2. Requirement Drift Findings

ID	Check	Severity	PRD Ref	File Path	Line(s)	Finding	Fix Instruction
RD-02-001	Logic Gate Fidelity — familiarity_score formula	HIGH	Artifact 23 §3 Threshold Constants	Execution /synthetic-data/clients.json	20, 63, 107, 200	scoring_constants documents SPP_FAMILIARITY_SCALE: "min(visit_count * 0.20, 0.60)" which caps familiarity at 0.60 for all visit counts ≥ 3. Artifact 23 §3 specifies familiarity_score = min(1.0, visit_count / FAMILIARITY_VISIT_CAP) where FAMILIARITY_VISIT_CAP=5. The formulas agree for visit_count 1–3 (both yield 0.20, 0.40, 0.60) but diverge at visit_count 4 (spec: 0.80, clients.json: 0.60 capped) and visit_count 5+ (spec: 1.0, clients.json: 0.60 capped). All current test assertion scores are correct because no scenario uses visit_count ≥ 4 — but if ECC implements the clients.json formula, carer selection will be wrong for established carer relationships. The P-7_note on line 63 (min(3 * 0.20, 0.60) = 0.60) further propagates the incorrect formula.	See FIX-002 — correct SPP_FAMILIARITY_SCALE in scoring_constants and update all P-7_note fields to use the correct formula.

ID	Check	Severity	PRD Ref	File Path	Line(s)	Finding	Fix Instruction
RD-04-001	Data Schema Conformance — P-8 field semantics	HIGH	Artifact 21 PRD §5; Artifact 23 §4 Gate 4	Execution /synthetic-data/clients.json	64, 108, 153, 201, 216, 261	P-8_do_not_send_flag is semantically inverted vs. the PRD. Artifact 21 PRD §5 defines P-8 as: "Binary carer flags — Per-carer flag: 'Client has accepted this carer before'" — a positive signal that floors familiarity_score at FAMILIARITY_BINARY_FLOOR=0.80 for that specific carer. clients.json labels it do_not_send_flag which implies a negative/exclusion signal — the opposite semantic. If ECC reads clients.json as the data model specification, it will implement P-8 as an exclusion filter rather than a familiarity floor, breaking matching logic for any client with an established preferred carer. Additionally, FAMILIARITY_BINARY_FLOOR=0.80 is absent from scoring_constants (line 15–34).	See FIX-003 — rename field, add constant, add test scenario.
RD-06-001	Detection Accuracy — P-8 FAMILIARITY_BINARY_FLOOR path	MEDIUM	Artifact 23 §3; Gate 4 P-8 branch	Execution /synthetic-data/incidents.json	all	No synthetic scenario exercises the P-8=true path. FAMILIARITY_BINARY_FLOOR=0.80 is a named threshold constant in Artifact 23 §3 but it is never tested. ECC could implement familiarity scoring correctly for the min(1.0, visit_count / FAMILIARITY_VISIT_CAP) formula and still mishandle the P-8 binary floor with no test failure.	Addressed by FIX-003 (add SYN-09 scenario with P-8=true carer). The scenario should verify that a carer with visit_count=1 but P-8=true achieves familiarity_score=0.80 (not 0.20).

3. Low-Severity Findings

ID	Check	Severity	File Path	Line(s)	Finding	Fix Instruction
IS-03-001	IS-03 Postcode Centroid Data Coverage	LOW	data/au_postcode_centroids_seed.csv	all (13 rows)	Seed file contains only the 13 postcodes referenced in clients.json. Proximity calculations for any client or carer outside these postcodes will fail with a missing-postcode error at runtime. Production deployment requires the full ABS Postal Areas ASGS Edition 3 dataset (~18,000 rows).	Acceptable for test harness. Before production: replace seed with full ABS dataset (CC BY 4.0). Download URL documented in file header. No ECC code change required — same schema.
IS-AUTH-001	SD-02 Auth Deferral Not Documented	LOW	Execution/Artifact_21_PRD.md	—	v1 auth deferral is architecturally sound (coordinator-desktop only, no public API) but is not explicitly stated as a documented design decision in the PRD. Without a documented deferral, v2 engineers may not know to add auth before exposing any API endpoint.	Add a single paragraph to PRD §8 Non-Functional Requirements: "Authentication and RBAC are deferred to v2. v1 operates as a coordinator-desktop tool with no public API endpoints. No multi-tenant access is provided. Auth must be specified before any API surface is exposed."
AX-02-001	AX-02 Env Var Not in CI Config	LOW	CI/CD configuration (not yet written)	—	HEMOCARE_AI_AX02_STATUS is correctly referenced in Artifact 25 §10 Gap 3 and incidents.json gap_resolutions.GLOBAL-AX02 — but there is no .env.example file or CI pipeline config documenting this variable alongside HEMOCARE_AI_TEST_MODE. An engineer running the test harness without this context will get exit_code=2 with no clear explanation.	Create .env.example with at minimum: HEMOCARE_AI_TEST_MODE=true, HEMOCARE_AI_AX02_STATUS=UNAVAILABLE (safe default for test runs), and a comment block referencing Artifact 25 §10 Gap 3.

4. Passed Checks

PASSED CHECKS

■	SD-01 (PHI in SMS)	All 4 template groups omit: client surname, full address, diagnosis, match score, composite_score, algorithm weights. CC-8 enforcement documented in config/sms_templates.py header.
■	SD-01 (PHI in audit log)	APPAuditLogEntry stores patient_id (UUID) only. No plaintext name, DOB, or condition in any log field. expected_audit.sql confirms no PHI literals in INSERT values.
■	SD-01B (Transition coverage SYN-01–SYN-05)	All 6 implemented scenarios generate complete, unbroken event chains in expected_audit.sql. SD-01B merge gate query structure is correct: "0 rows = PASS".
■	SD-01B (Hash chain schema) entry_hash (SHA-256) and previous_hash fields present in	synthetic_expected_audit_events DDL. Hash-chain integrity is structural, not procedural.
■	SD-01B (7-year retention)	AUDIT_LOG_RETENTION_YEARS=7 in scoring_constants. Write-once S3 + CloudTrail architecture confirmed in Artifact 23 §9 Infrastructure Requirements.
■	SD-01B (Append-only)	Immutable store architecture: no UPDATE/DELETE path exists in the Logic Spec for any audit log entry.
■	SD-02 (v1 scope)	Auth deferral architecturally consistent with coordinator- desktop v1. No multi-tenant access, no public API. Documented as LOW gap (IS-AUTH-001) not security failure.
■	SD-03 (CRIT-02 satisfied)	config/sms_templates.py uses template.replace() only. No f-strings with user data, no format() with unvalidated input, no LLM in any SMS rendering path.
■	SD-04 (RAG isolation)	Not applicable — v1 is rule-based. No vector store, no embeddings pipeline, no RAG component.
■	SD-05 (Secrets)	Zero hardcoded credentials in any file. All external service keys (AWS, MessageMedia/SNS, AlayaCare) are environment variable references in Artifact 23 §9.
■	SD-06 (Input validation)	get_template() raises ValueError on invalid key or None template (ACT_P_01_VARIANT_C unreachable guard). Template rendering is safe by construction (CRIT-02).
■	RD-01 (Feature completeness) All 10 user stories (US-01 through US-10, Artifact 24)	are traceable to named Logic Spec gates in expected_audit.sql event chains.
■	RD-02 (Threshold constants) FAMILIARITY_VISIT_CAP=5, PROXIMITY_MAX_KM=25,	SHORTLIST_MAX_CANDIDATES=3, SPP_COMPLETENESS_THRESHOLD=0.80 all present in scoring_constants and correctly applied in all 6 scenario assertion scores. (Formula constant is correct; documentation string is wrong — see RD-02-001.)

■	RD-03 (HITL checkpoint)	Gate 7 (COORDINATOR_APPROVED) correctly gates all Level 3 actions: carer notification (ACT_C_01), briefing (ACT_C_02), client notification (ACT_P_01), family notification (ACT_F_01). No notification fires before COORDINATOR_APPROVED event.
■	RD-04 (APP extension schema) APPAuditLogEntry correctly extended with:	cross_border_disclosure, app8_basis, consent_record_id, consent_version, guard_id, guard_passed, guard_block_reason. All APP-specific fields present in expected_audit.sql DDL.
■	RD-05 (Output format)	TEMPLATES dispatch dict correctly keyed for all 10 template variants. assert_copy_approved() advisory mode (HOMECARE_AI_TEST_MODE) correctly implemented. VERSION logged for audit traceability.

5. Agent Fix Queue

Ordered CRITICAL → HIGH. Execute in sequence — each fix may affect test assertion counts.

FIX-001 [CRITICAL]

SD-01B-001

Files: Execution/synthetic-data/clients.json, Execution/synthetic-data/incidents.json, Execution/synthetic-data/expected_audit.sql

Action: Add three synthetic scenarios to close coverage gaps for canonical events COORDINATOR_OVERRIDE, E3_GATE_BLOCKED, and DUPLICATE_VACANCY_SUPPRESSED. Additionally close EC-03 partial SPP path (see FIX-003 note).

Scenario SYN-06 — COORDINATOR_OVERRIDE (Gate 7b)

Coordinator receives shortlist, overrides system ranking, selects the 2nd-ranked carer (not top-ranked). The COORDINATOR_OVERRIDE event must fire before CARER_NOTIFIED. Demonstrates that the system logs the override and proceeds with the coordinator's chosen carer — not the highest-scoring candidate.

Expected events: VACANCY_DETECTED → SHORTLIST_GENERATED →
HITL_ESCALATED → COORDINATOR_OVERRIDE → CARER_NOTIFIED →
BRIEFING_SENT → CLIENT_NOTIFIED → FAMILY_NOTIFIED → RESOLVED

Add to clients.json: new client record with scenario="SYN-06" Add to incidents.json: incident with coordinator_selection_rank: 2 (overrides rank 1) Add to expected_audit.sql: 9 APPAuditLogEntry INSERTs with guard_id= 'G-7B' on COORDINATOR_OVERRIDE event

Scenario SYN-07 — E3_GATE_BLOCKED

Family notification attempted but client_notified=false at time of family send attempt. Gate 12 E-3 guard fires E3_GATE_BLOCKED. The incident must halt at CLIENT_NOTIFIED state — family notification is not sent. Demonstrates the E-3 ordering constraint.

Expected events: VACANCY_DETECTED → ... → CARER_NOTIFIED →
BRIEFING_SENT → E3_GATE_BLOCKED (family send rejected) → [alert to coordinator]

Add to clients.json: client record with scenario="SYN-07" Add to incidents.json: incident with force_e3_violation: true in test pre-conditions Add to expected_audit.sql: APPAuditLogEntry INSERT for E3_GATE_BLOCKED with guard_passed=false, guard_block_reason='client_notified=false at Gate 12 execution'

Scenario SYN-08 — DUPLIC ATE_VACANCY_SUPPRESS ED (Gate 0)

A second VACANCY_DETECTED event fires for a vacancy ID that already exists in state SHORTLIST_READY or later. Gate 0 idempotency guard fires DUPLICATE_VACANCY_SUPPRESSED. Processing halts — no duplicate shortlist is generated.

Expected events: DUPLICATE_VACANCY_SUPPRESSED (terminal – single event)

Add to clients.json: reuse existing client, add scenario="SYN-08" section Add to incidents.json: incident with duplicate_of: "SYN-01-incident-id" in pre_conditions Add to expected_audit.sql: 1 APPAuditLogEntry INSERT for DUPLICATE_VACANCY_SUPPRESSED with state_before='SHORTLIST_READY', state_after='SHORTLIST_READY'

FIX-002 [HIGH]

RD-02-001

File: Execution/synthetic-data/clients.json

Action: Correct the documented familiarity score formula in scoring_constants and all P-7_note fields.

```
BEFORE (line 20):
  "SPP_FAMILIARITY_SCALE": "min(visit_count * 0.20, 0.60)"

AFTER (line 20):
  "SPP_FAMILIARITY_SCALE": "min(1.0, visit_count / FAMILIARITY_VISIT_CAP)"

Also add to scoring_constants (after line 20):
  "FAMILIARITY_VISIT_CAP": 5
```

Update P-7_note fields to use the correct formula:

```
BEFORE (line 63):
  "P-7_note": "David Okafor has 3 confirmed prior visits. familiarity_score = min(3 * 0.20, 0.60) = 0.60"

AFTER (line 63):
  "P-7_note": "David Okafor has 3 confirmed prior visits. familiarity_score = min(1.0, 3/5) = 0.60"
```

Apply same correction to P-7_note on lines 107 and 200. Note: computed scores in expected_shortlist blocks are unaffected — all current test scenarios use visit_count ≤ 3, so the numeric outputs do not change. Only the documentation string changes.

FIX-003 [HIGH]

RD-04-001 + RD-06-001

File: Execution/synthetic-data/clients.json, Execution/synthetic-data/incidents.json, Execution/synthetic-data/expected_audit.sql

Step 1 — Correct field name (clients.json lines 64, 108, 153, 201, 216, 261)

```
BEFORE: "P-8_do_not_send_flag": null
AFTER: "P-8_binary_acceptance_flag": {}
```

P-8 is a **dict** keyed by carer_id (same structure as P-7_visit_history), not a scalar null. Value structure: { "carer_id": true } means "client has explicitly accepted this carer before — floor familiarity at FAMILIARITY_BINARY_FLOOR=0.80 for this carer."

Step 2 — Add FAMILIARITY_BINARY_FLOOR to scoring_constants (clients.json line 20 block)

```
Add after FAMILIARITY_VISIT_CAP:
  "FAMILIARITY_BINARY_FLOOR": 0.80
```

Step 3 — Add SYN-09 scenario (new entries in clients.json, incidents.json, expected_audit.sql)

SYN-09 tests the P-8 binary acceptance floor: client has P-8_binary_acceptance_flag: { "ca3e0001-...": true } for a carer with only visit_count=1 (which would normally yield familiarity_score=0.20). The SHORTLIST_GENERATED event must show this carer with familiarity_score=0.80 (floor applied), not 0.20. Assert in expected_audit.sql:

```
threshold_applied = 'FAMILIARITY_BINARY_FLOOR=0.80 (P-8_binary_acceptance_flag=true)'
```

FIX-004 [HIGH]

CP-01-001

File: config/sms_templates.py, lines 45–46

Type: Process action — NOT an ECC code change

```
Current state (lines 45-46):
  APPROVED_BY = None # ← BLOCKS go-live
  APPROVED_DATE = None # ← BLOCKS go-live

Required state (set by Privacy Counsel after review):
  APPROVED_BY = "Privacy Counsel – [Full Name]"
  APPROVED_DATE = "YYYY-MM-DD" # ISO 8601
```

Additionally (lines 49–50), set on coordinator clarity review:

```
COORDINATOR_PROXY_REVIEW_BY = "Angela CC-001"  
COORDINATOR_PROXY_REVIEW_DATE = "YYYY-MM-DD"
```

Templates requiring review (4 groups)

- ACT_C_01 — Carer assignment SMS (Gate 8): no PHI, 133 chars max
- ACT_C_02_VARIANT_A/B/C — Carer briefing SMS (Gate 9): includes client first name + full address (released post-confirmation only)
- ACT_P_01_VARIANT_A/B — Client notification SMS (Gate 11): no surname, no SPP reference
- ACT_F_01_VARIANT_A/B — Family notification SMS (Gate 12): client first name only, pronoun-safe

Dependency chain: CP-01 unblocks production go-live. Test harness is unaffected (HOMECARE_AI_TEST_MODE=true bypasses assert_copy_approved() with WARNING log). No code change needed — gate mechanism is correctly implemented.

6. Regression Test Recommendations

For each CRITICAL and HIGH fix, the following tests must pass before the SD-01B merge gate is re-run.

TEST-001 [after FIX-001 / SYN-06]: Assert that when coordinator_selection_rank=2, the COORDINATOR_OVERRIDE event fires with action_taken containing the overridden carer_id, and CARER_NOTIFIED fires with the rank-2 carer – not rank-1.

TEST-002 [after FIX-001 / SYN-07]: Assert that when client_notified=false at Gate 12 execution, E3_GATE_BLOCKED fires and no ACT_F_01 template is rendered. The family contact must receive no SMS.

TEST-003 [after FIX-001 / SYN-08]: Assert that a second VACANCY_DETECTED with an existing vacancy_id fires DUPLICATE_VACANCY_SUPPRESSED and does not advance state or generate a second shortlist.

TEST-004 [after FIX-002]: Unit test familiarity_score for visit_count values 0, 1, 2, 3, 4, 5, 6:

- visit_count=0 → 0.0
- visit_count=1 → 0.20
- visit_count=3 → 0.60
- visit_count=4 → 0.80 ← key regression case (was capped at 0.60 with wrong formula)
- visit_count=5 → 1.0
- visit_count=6 → 1.0 ← capped at 1.0

TEST-005 [after FIX-003]: Unit test P-8 binary floor:

- carer with visit_count=1, P-8_binary_acceptance_flag[carer_id]=true
→ familiarity_score = 0.80 (floor applied)
- carer with visit_count=1, no P-8 entry
→ familiarity_score = 0.20 (normal formula)

TEST-006 [after FIX-003 / SYN-09]: Integration test asserting SHORTLIST_GENERATED event for SYN-09 has threshold_applied containing 'FAMILIARITY_BINARY_FLOOR=0.80' for the P-8=true carer.

TEST-007 [SD-01B merge gate re-run]: After FIX-001, re-run the SD-01B merge gate query in expected_audit.sql. Must return 0 rows. Any row returned = CRITICAL, blocks merge.

7. Pre-Sprint Gate Status

Findings in this audit that have external dependencies (not resolvable by ECC alone):

Gate	Status	Owner	Due	Blocks
AX-01 — P-9 absent from v1 schema	CONFIRMED ■	Engineer	—	clients.json header, line 6
AX-02 — AlayaCare write-back confirmation	PENDING ■	AlayaCare integration lead	2026-04-01	Production deploy
SC-07 — Google Maps DPA vs APP 8	PENDING ■	Legal / Privacy Counsel	2026-04-01	Full ABS dataset replacement
B1-B — SMS Copy Approval	PENDING ■	Privacy Counsel + Angela CC-001	TBD	Production deploy (FIX-004)
Sketch-gate — Coordinator card wireframe	PENDING ■	Designer	TBD	US-10 build

ECC may build against the 6 existing synthetic scenarios immediately. All external gates are non-blocking for the development build. They are blocking for production deploy only.

8. Spec Coverage Map

Summary of what the synthetic test scenarios cover vs. the full Logic Spec event registry:

Canonical Event	Artifact 23 §8	Covered by Synthetic	Scenario
VACANCY_DETECTED	Gate 1	■	SYN-01 through SYN-05
SHORTLIST_GENERATED	Gate 6	■	SYN-01, 02, 04a, 04b, 05
VACANCY_UNRESOLVED (no candidates)	Gate 4 terminal	■	SYN-03
HITL_ESCALATED (coordinator approval)	Gate 7	■	SYN-01, 02, 04a, 04b, 05
COORDINATOR_APPROVED	Gate 7a	■	SYN-01, 02, 04a, 04b, 05
COORDINATOR_OVERRIDE	Gate 7b	■	FIX-001 → SYN-06
CARER_NOTIFIED	Gate 8	■	SYN-01, 02, 04a, 04b
CARER_CONFIRMED	Gate 10	■	SYN-01, 02, 04a
CARER_DECLINED	Gate 10 declined	■	SYN-04b
BRIEFING_SENT	Gate 9	■	SYN-01, 02, 04a
CLIENT_NOTIFIED	Gate 11	■	SYN-01, 02
FAMILY_NOTIFIED	Gate 12	■	SYN-01, 02
E3_GATE_BLOCKED	Gate 12 guard	■	FIX-001 → SYN-07
RESOLVED	Gate 13	■	SYN-01
VACANCY_UNRESOLVED (all declined)	Gate 14	■	SYN-05
DUPLICATE_VACANCY_SUPPRESSED	Gate 0	■	FIX-001 → SYN-08
CARER_NOTIFICATION_FAILED	Gate 8 failure	■	Not yet scheduled
CLIENT_NOTIFICATION_UNAVAILABLE	Gate 11 failure	■	Not yet scheduled
P-8 FAMILIARITY_BINARY_FLOOR path	Gate 4 P-8 branch	■	FIX-003 → SYN-09
EC-04 cold-start (P-10=0.0)	Briefing MINIMAL	■	SYN-05
EC-03 partial SPP (0 < P-10 < 0.80)	Briefing partial	■	Not yet scheduled
IS-03 postcode centroid fallback	Proximity gate	■	All scenarios (SC07=false)
AX-02 write-back gate	Post-RESOLVE D	■ (advisory)	Asserted in GLOBAL-AX02
SMS advisory body assertions	B1-B gate	■ (advisory)	SYN-01-A9 through A12

Coverage rate: 15 / 24 distinct paths = 62.5% before FIX-001 through FIX-003. After fixes: 19 / 24 = 79.2%. Remaining 5 gaps (CARER_NOTIFICATION_FAILED, CLIENT_NOTIFICATION_UNAVAILABLE, EC-03, and 2 others) are lower-priority and may be deferred to a v1.1 test sprint.

Artifact 26 produced by: Harness Audit Grader (pm-execution skill)

Audit inputs: Artifact 21 (PRD), Artifact 23 (Logic Spec), Artifact 24 (User Stories), Artifact 25 (Synthetic PHI Generator), config/sms_templates.py, data/au_postcode_centroids_seed.csv

Next action: ECC executes FIX-001 through FIX-003. FIX-004 is routed to Privacy Counsel.

Re-audit required after fixes before SD-01B merge gate is cleared.